

A Three-Dimensional Counterexample to a Split-Ideal Question

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Abstract

For a standard graded, locally finite, semiconnected algebra $A_\bullet$, Dramburg asked whether every two-sided ideal J satisfying $\dim_k(A/J) = \dim_k A_0$ must be split, meaning that the natural map $A_0 \rightarrow A/J$ is an isomorphism. We give a negative answer under exactly these hypotheses. Over every field, the direct product of the field with the algebra of dual numbers has a three-dimensional grading and a one-dimensional two-sided ideal for which the required dimensions agree, but the natural map has a nonzero kernel.

Keywords. graded algebra; semiconnected algebra; split ideal; dual numbers.

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1 The question and the counterexample

Let k be a field. A nonnegatively graded k -algebra $A_\bullet = \bigoplus_{i \geq 0} A_i$ is called *standard graded* if it is generated by $A_0 \cup A_1$, *locally finite* if every A_i is finite-dimensional over k , and *semiconnected* if A_0 is semisimple. A possibly ungraded two-sided ideal $J \triangleleft A$ is *split* with respect to this grading if the composite

$$A_0 \hookrightarrow A \twoheadrightarrow A/J$$

is an isomorphism. These definitions and the question of whether $\dim_k(A/J) = \dim_k A_0$ forces splitness appear in [Dra26, Definitions 1.1 and 4.1, Question 6.1].

Theorem 1. *For every field k , there is a three-dimensional standard graded, locally finite, semiconnected unital k -algebra $A_\bullet$ and a two-sided ideal $J \triangleleft A$ such that*

$$\dim_k(A/J) = \dim_k A_0,$$

while the natural map $A_0 \rightarrow A/J$ is not an isomorphism. Consequently, the codimension condition in Question 6.1 of [Dra26] does not, by itself, force an ideal to be split.

Proof. Let $R = k[\varepsilon]/(\varepsilon^2)$ and take the direct-product algebra

$$A = k \times R.$$

It is useful to write

$$e = (1, 0), \quad f = (0, 1), \quad t = (0, \varepsilon).$$

Then $1_A = e + f$, the elements e, f, t form a k -basis of A , and direct-product multiplication gives

$$e^2 = e, \quad f^2 = f, \quad ef = fe = 0, \quad et = te = 0, \quad ft = tf = t, \quad t^2 = 0. \quad (1)$$

Define a grading by

$$A_0 = ke \oplus kf, \quad A_1 = kt, \quad A_i = 0 \quad (i \geq 2).$$

The relations in (1) show that $A_i A_j \subseteq A_{i+j}$ in every degree, and $1_A \in A_0$. Moreover, $A = A_0 \oplus A_1$, so $A_0 \cup A_1$ generates A . The homogeneous dimensions are $2, 1, 0, 0, \dots$, and $A_0 \cong k \times k$ is semisimple. Hence this grading is standard, locally finite, and semiconnected.

Now set

$$J = ke = k \times 0.$$

For $x = ae + bf + ct \in A$ and $de \in J$, (1) gives $x(de) = (ad)e = (de)x$. Thus J is a two-sided ideal. Projection onto the second direct-product factor induces an isomorphism

$$A/J \xrightarrow{\sim} R, \quad (a, b + c\varepsilon) + J \mapsto b + c\varepsilon.$$

It follows that $\dim_k(A/J) = 2 = \dim_k A_0$.

Under this identification, the natural map on degree zero is

$$A_0 \longrightarrow R, \quad ae + bf \mapsto b.$$

Its kernel is $ke = J \cap A_0$, which is nonzero over every field. The map is therefore not injective and cannot be an isomorphism. Thus J is not split. $\square$

Remark 1 (Scope). *The construction is characteristic-free, and zero homogeneous pieces in degrees at least two are allowed by the stated hypotheses. It exploits a decomposable degree-zero algebra and makes no claim that dimension three is minimal. Dramburg proves splitness for ideals of the special form arising from an ungraded isomorphism of graded algebras [Dra26, Proposition 4.7]; Theorem 1 concerns an arbitrary two-sided ideal under the codimension condition alone and does not contradict that result.*

Computational and AI assistance disclosure

The construction, proof verification, and preparation of this note used OpenAI Codex as a mathematical and writing assistant. No numerical or symbolic computation is used in the proof: every assertion follows from the displayed multiplication table.

References

- [Dra26] Darius Dramburg. Isomorphisms of graded semiconnected algebras, 2026. arXiv:2609.03288v1.